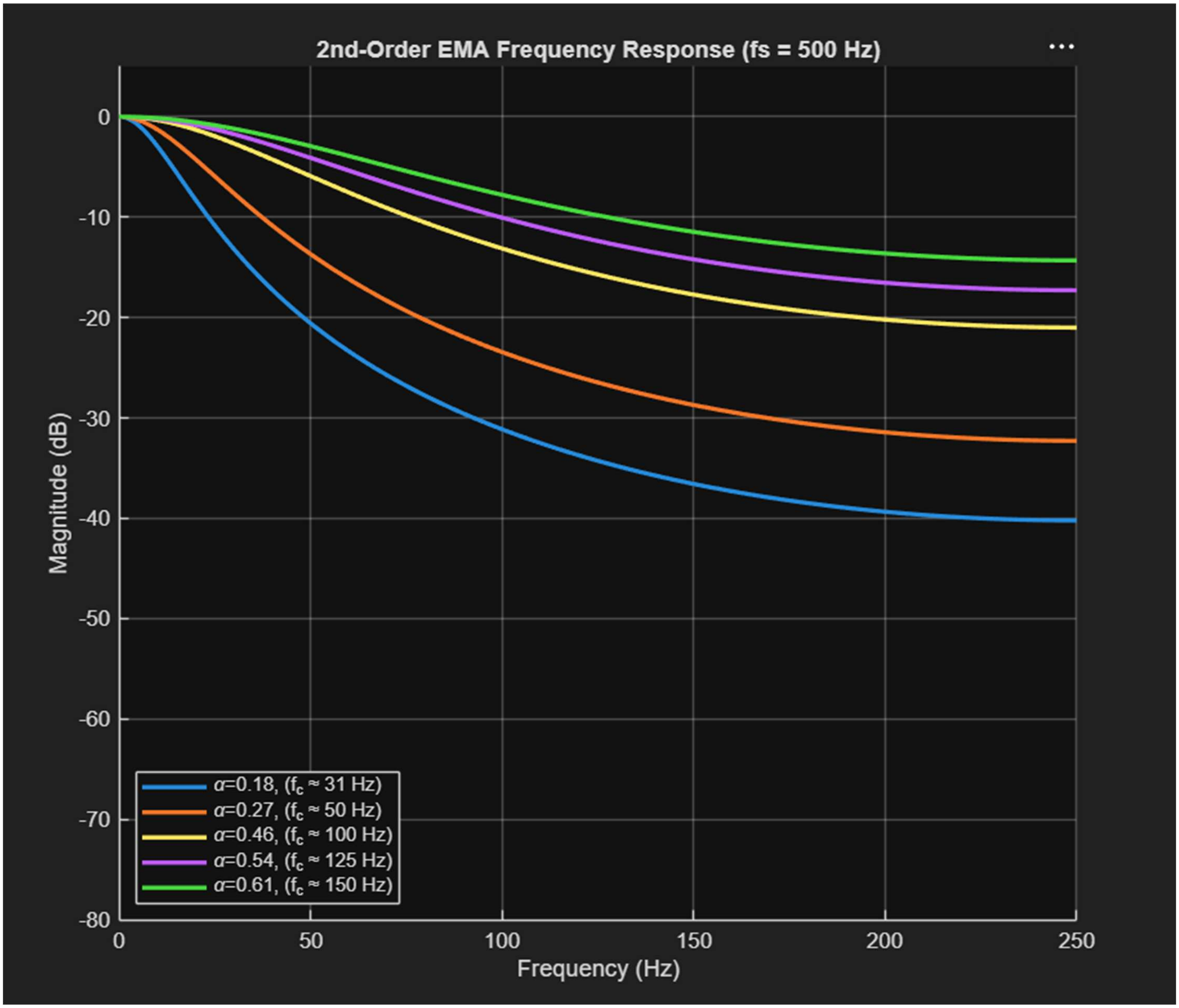
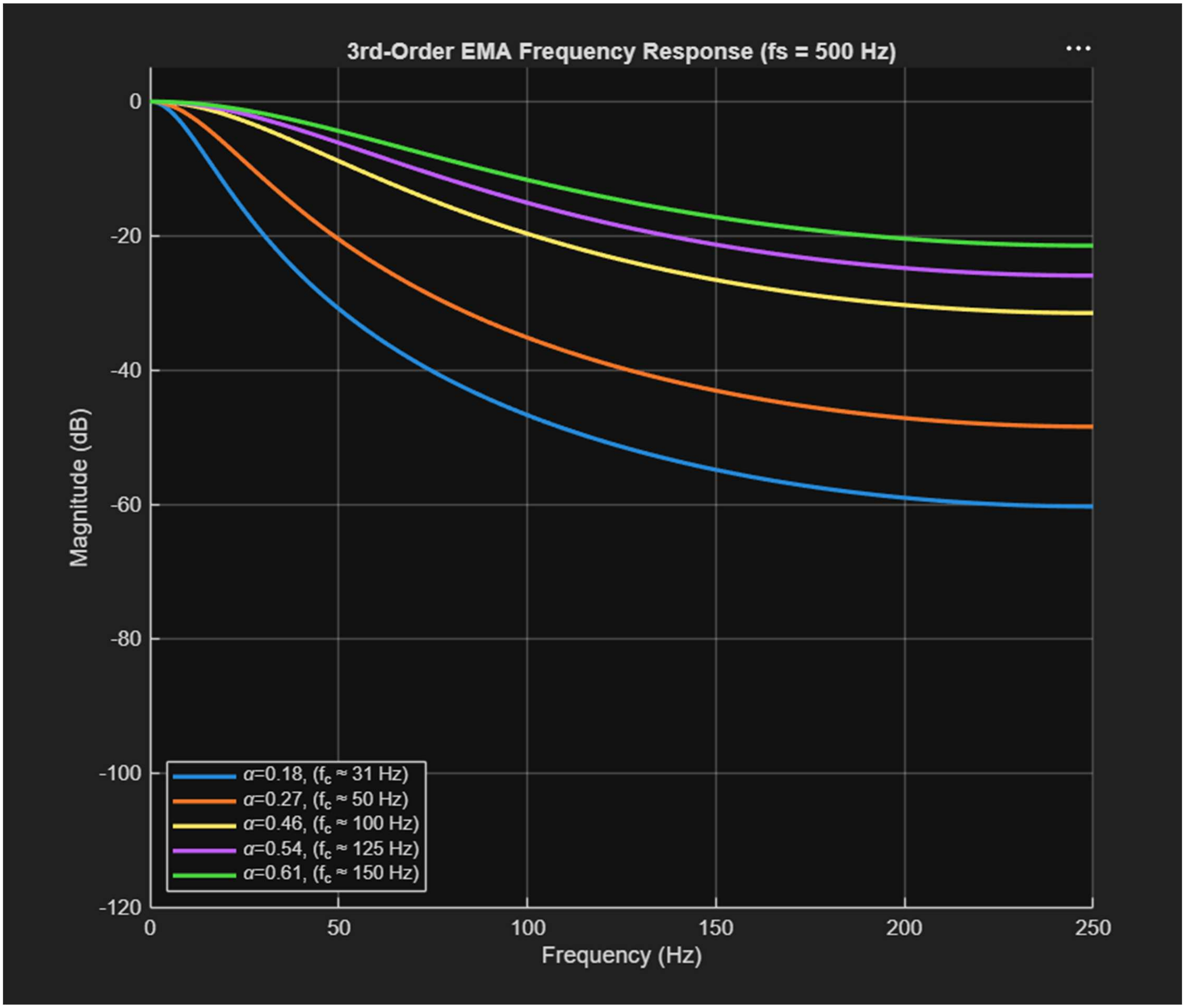
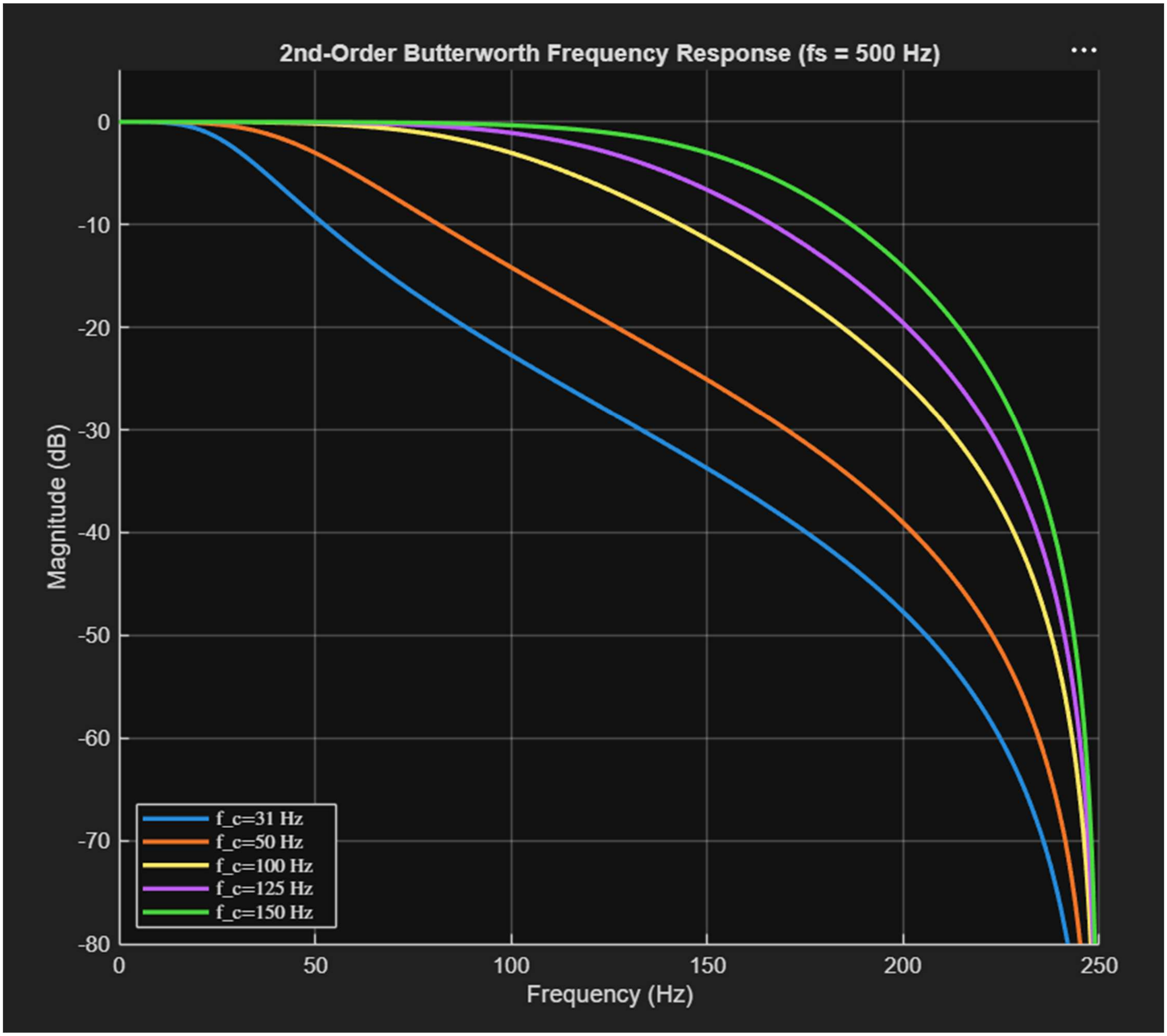
EMA 2nd Order Digital Filter.

Matlab frequency responses:





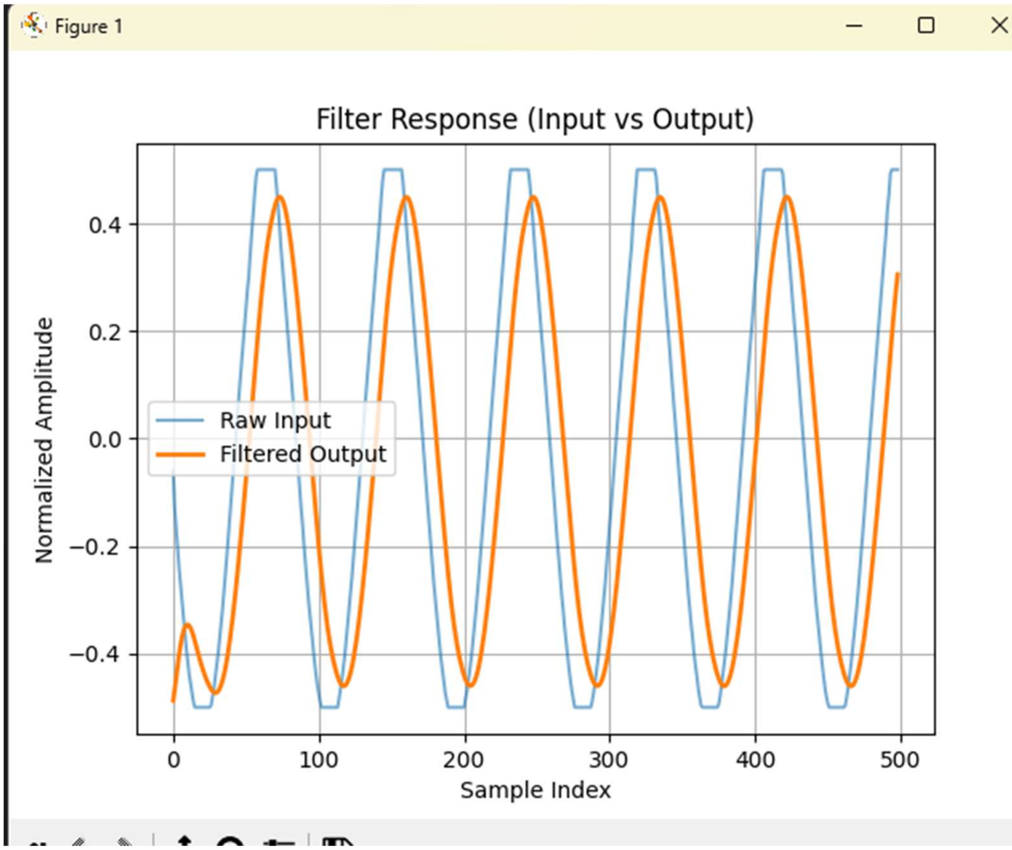
3rd-Order EMA Frequency Response (fs = 500 Hz)
Magnitude (dB)
Frequency (Hz)
0
-20
-40
-60
-80
-100
-120
0
50
100
150
200
250
α=0.18, (fc ≈ 31 Hz)
α=0.27, (fc ≈ 50 Hz)
α=0.46, (fc ≈ 100 Hz)
α=0.54, (fc ≈ 125 Hz)
α=0.61, (fc ≈ 150 Hz)


2nd-Order Butterworth Frequency Response (fs = 500 Hz)
Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
-70
-80
Frequency (Hz)
0
50
100
150
200
250
f_c=31 Hz
f_c=50 Hz
f_c=100 Hz
f_c=125 Hz
f_c=150 Hz

Corresponds to EMA_And_Accel_Logging

Plotted with serialLogAndPlot.py

50 Hz input, 2nd ord, 30 Hz cutoff



100 Hz

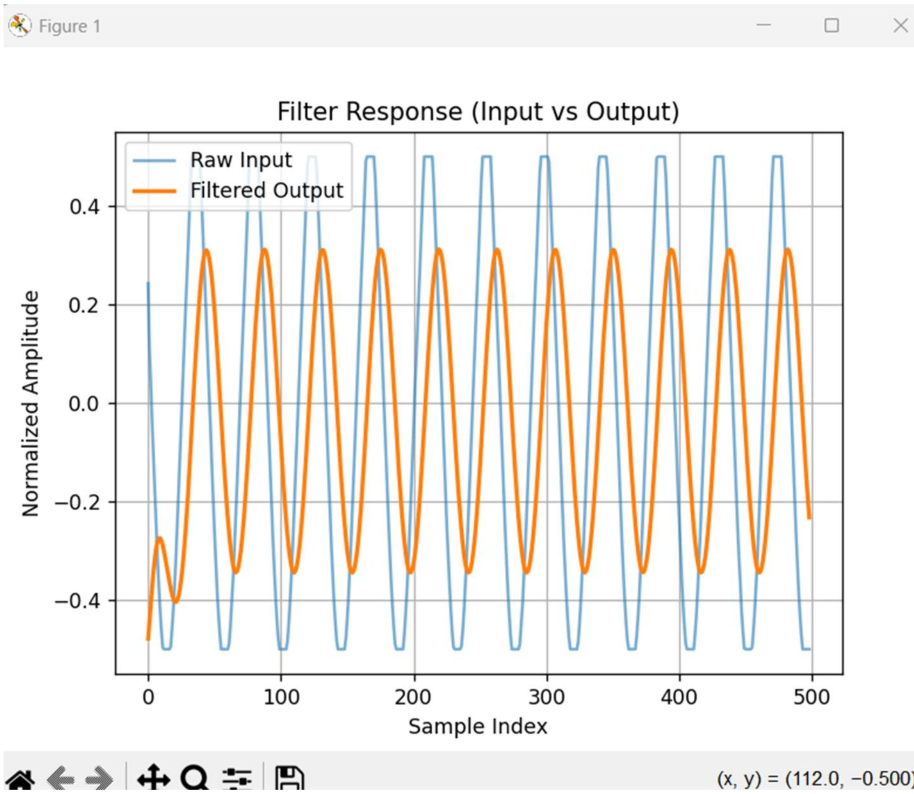


Figure 1
Filter Response (Input vs Output)
Raw Input
Filtered Output
0.4
0.2
0.0
−0.2
−0.4
Normalized Amplitude
0
100
200
300
400
500
Sample Index
(x, y) = (112.0, −0.500)

200 Hz

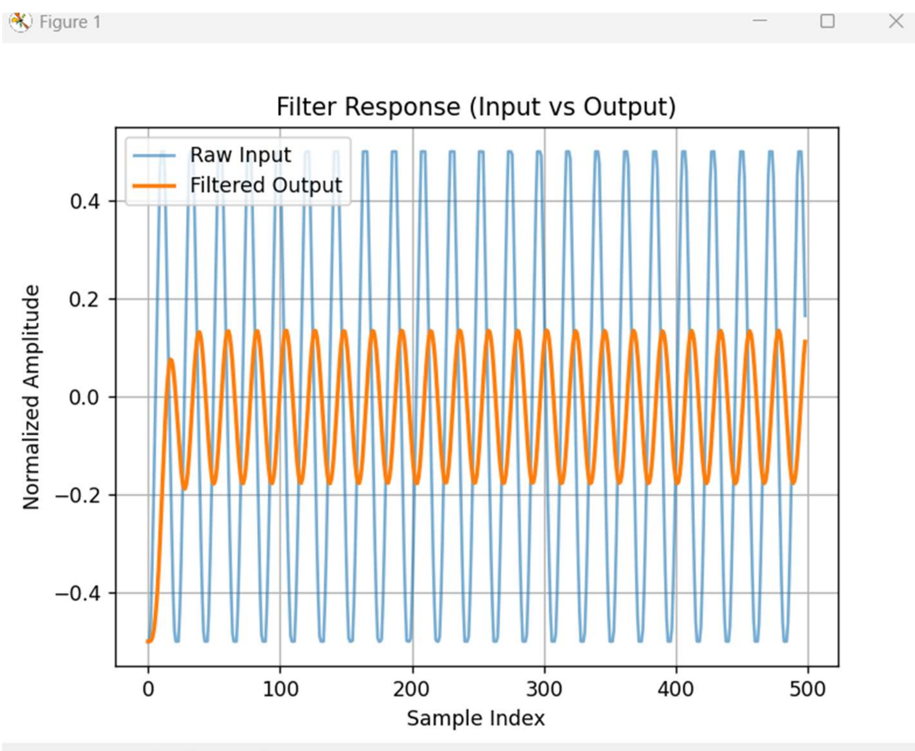
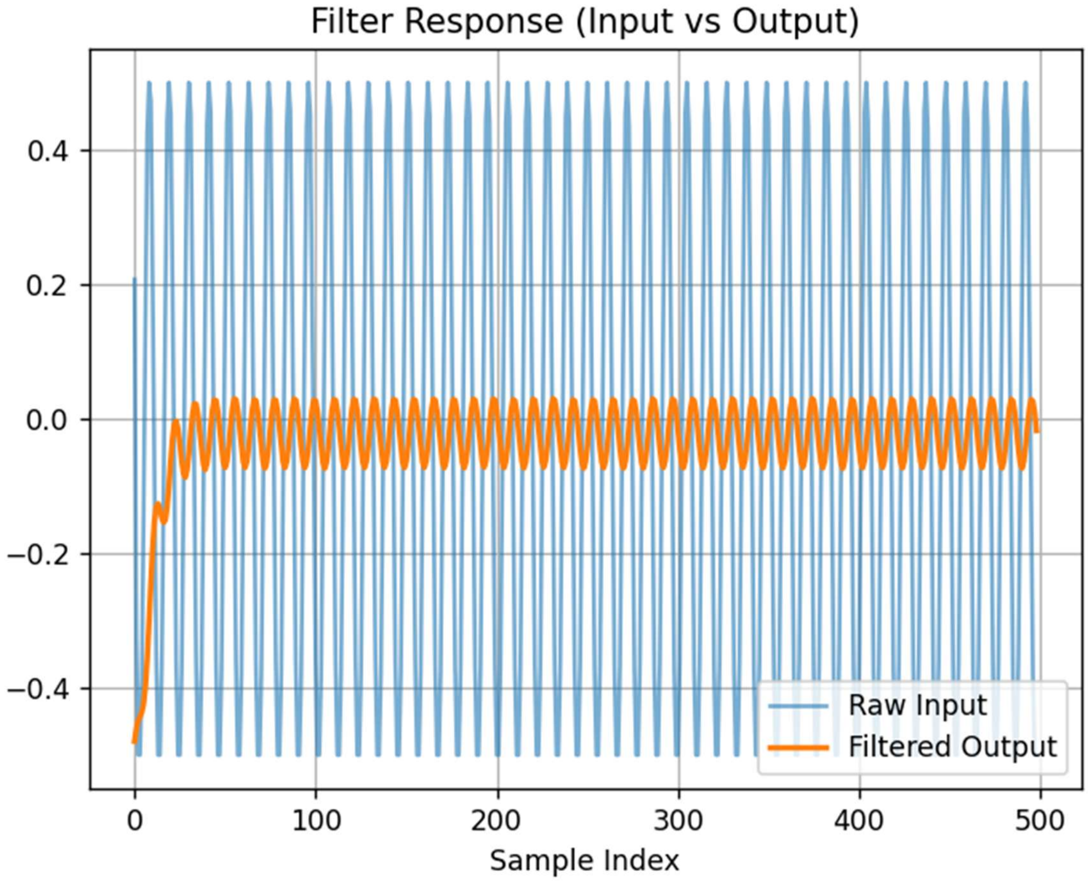
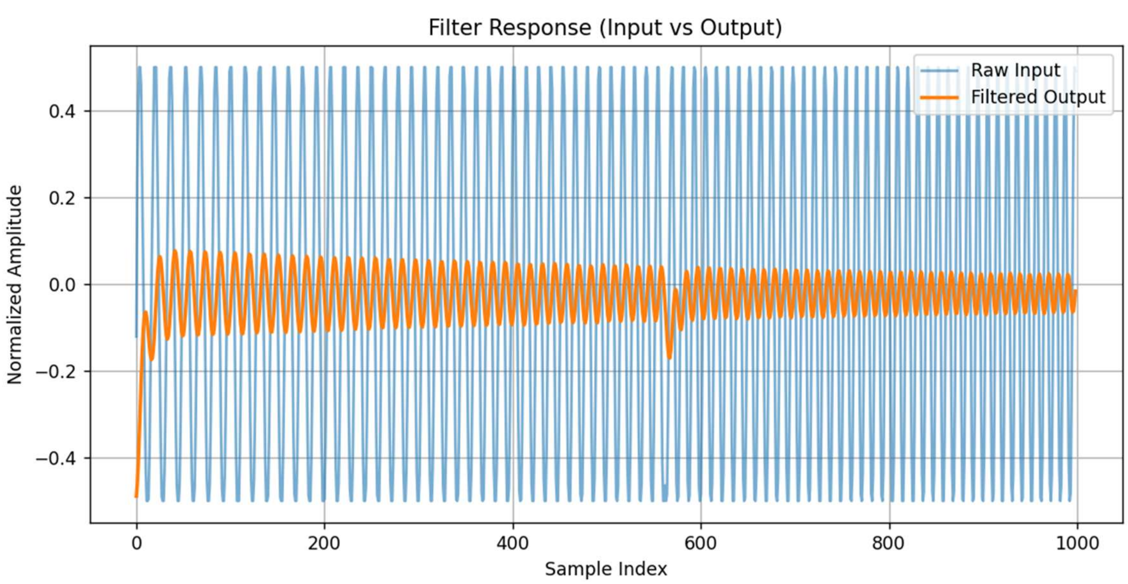


Figure 1
Filter Response (Input vs Output)
Raw Input
Filtered Output
0.4
0.2
0.0
−0.2
−0.4
Normalized Amplitude
0
100
200
300
400
500
Sample Index

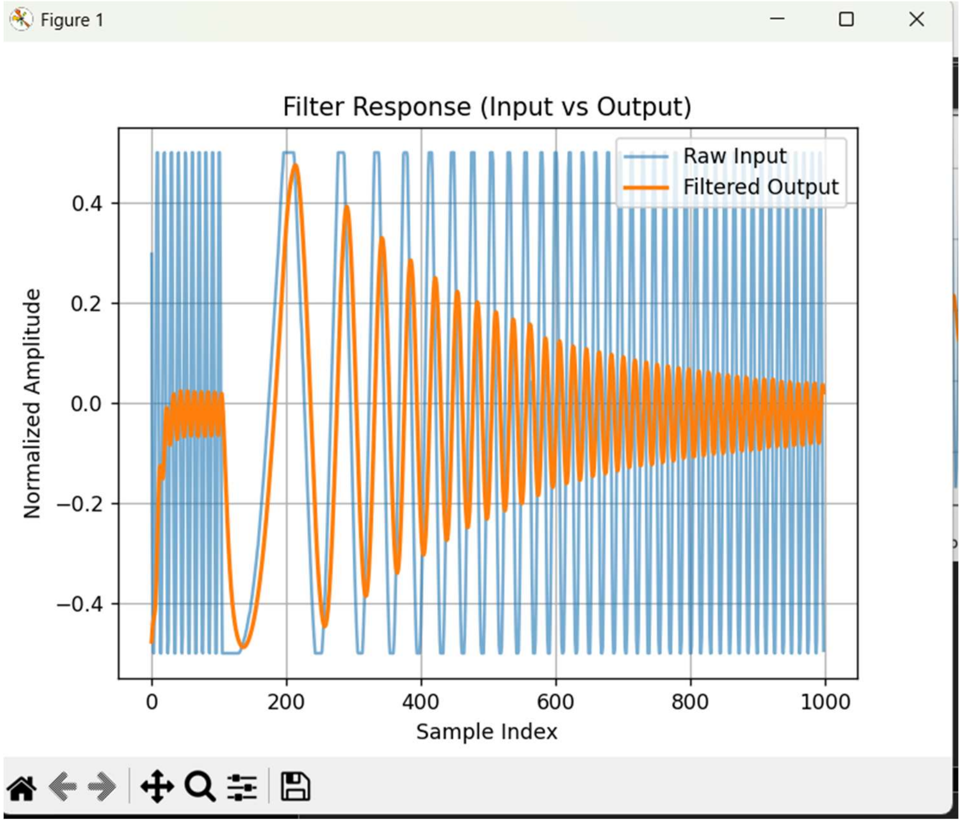
400 Hz



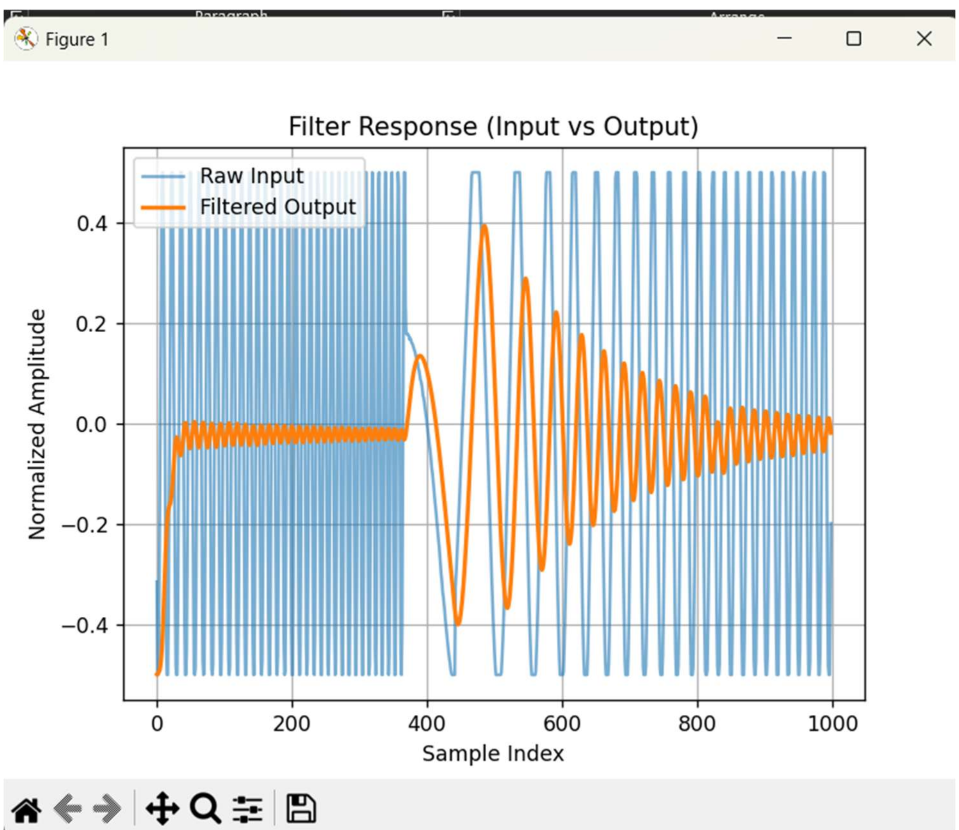
Sweep: 1-450 Hz, 500 ms



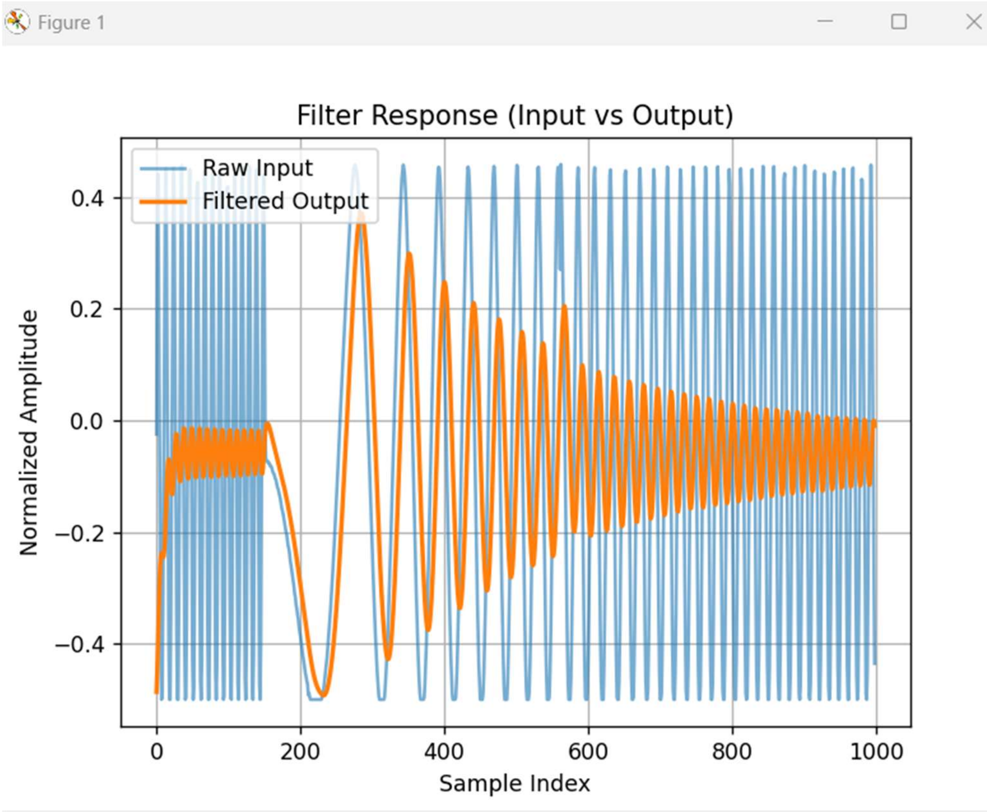
1-450 Hz, 250 ms sweep. 3.3 amp, 1.650 offset



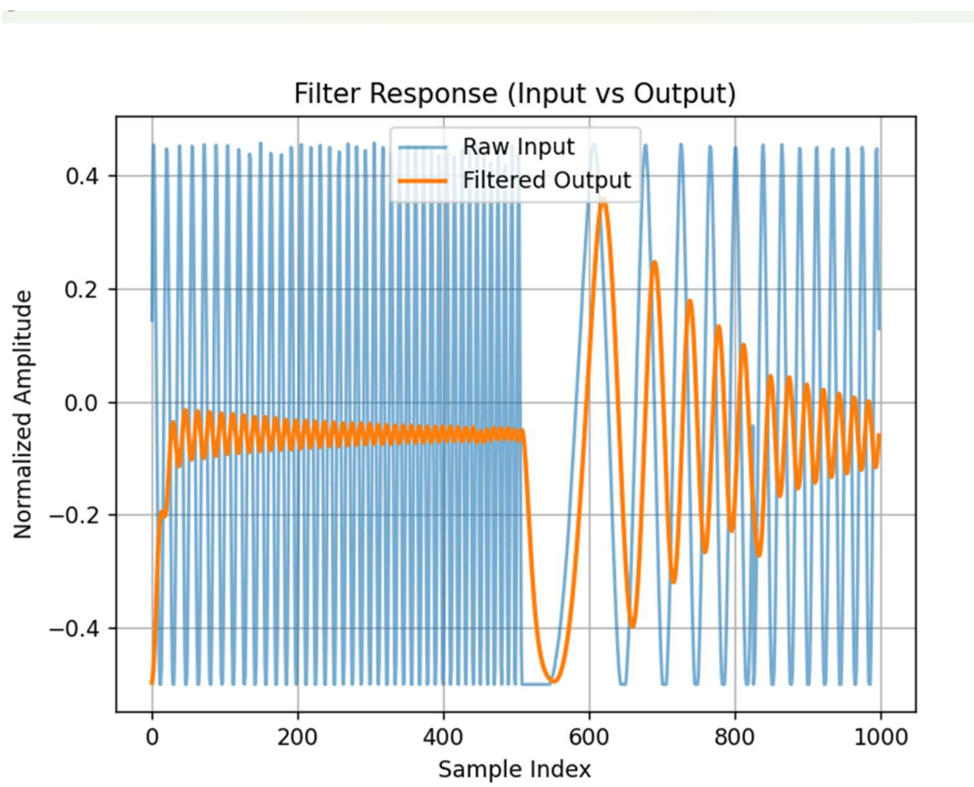
3rd order. 1-450 Hz sweep, 250 ms sweep. 3.3 amp, 1.650 offset



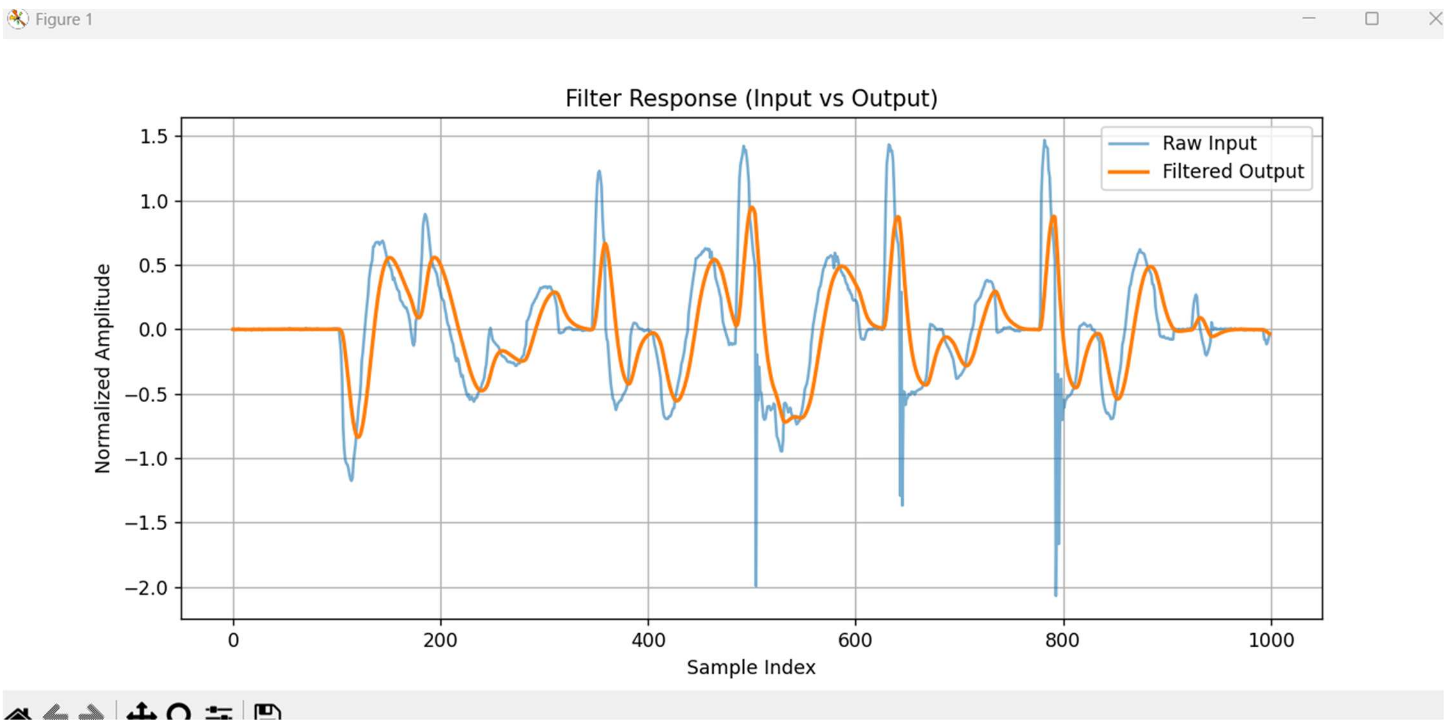
1-450 Hz, 250 ms sweep. 3.0 V amp, 1.550 V offset



3rd order. 1-450 Hz sweep, 250 ms sweep. 3.0 V amp, 1.550 offset

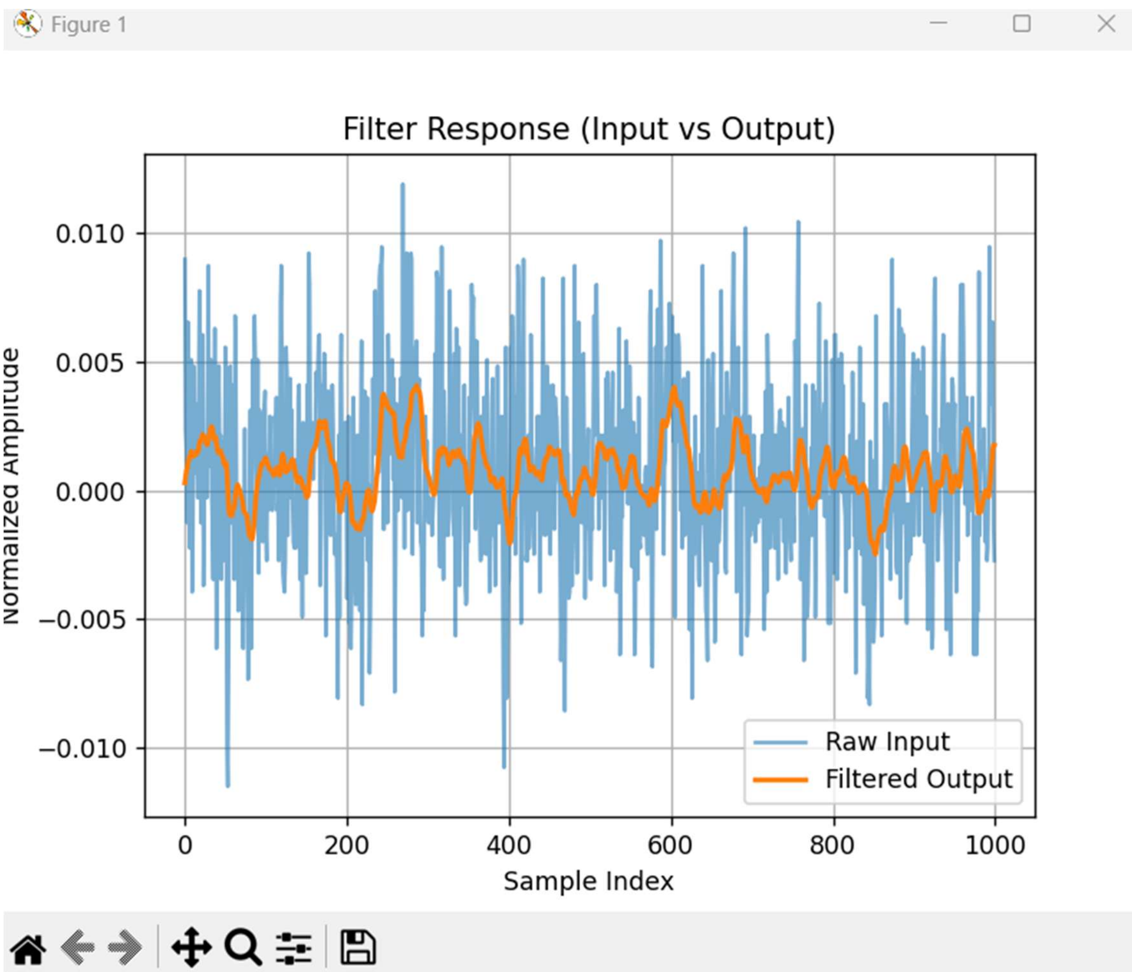


Accelerometer Raw vs filtered accelerometer data (x axis):



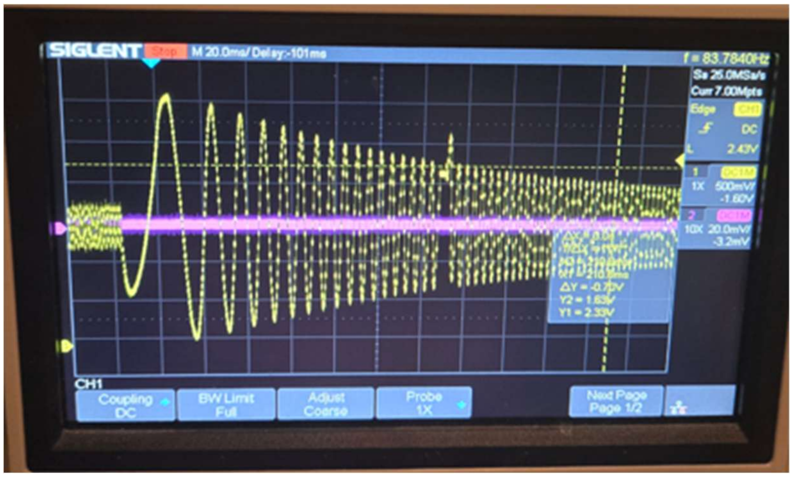
Timer Based version of the EMA, cutoff = 31 Hz, second order. Corresponds to EMA_And_Accel_Logging_Timer.py

No movement, just noise measurement:

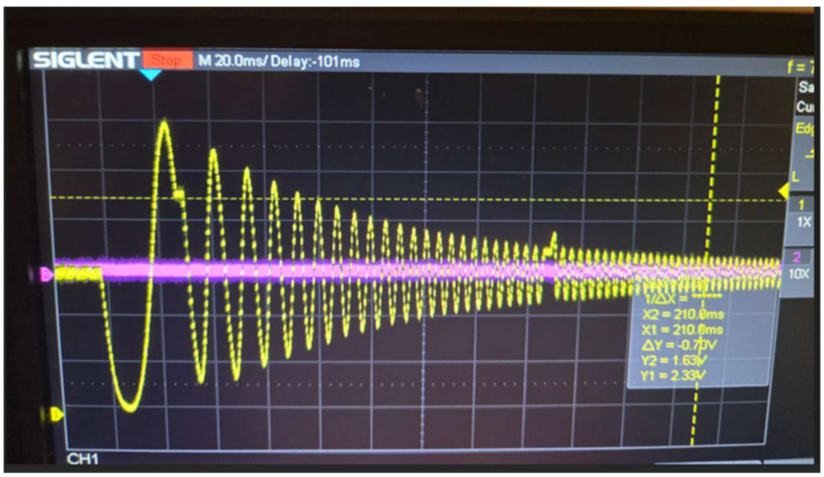


Scope Photos : ADC -> EMA -> DAC -> SCOPE, 3.0 V peak 1-1kHz sweep

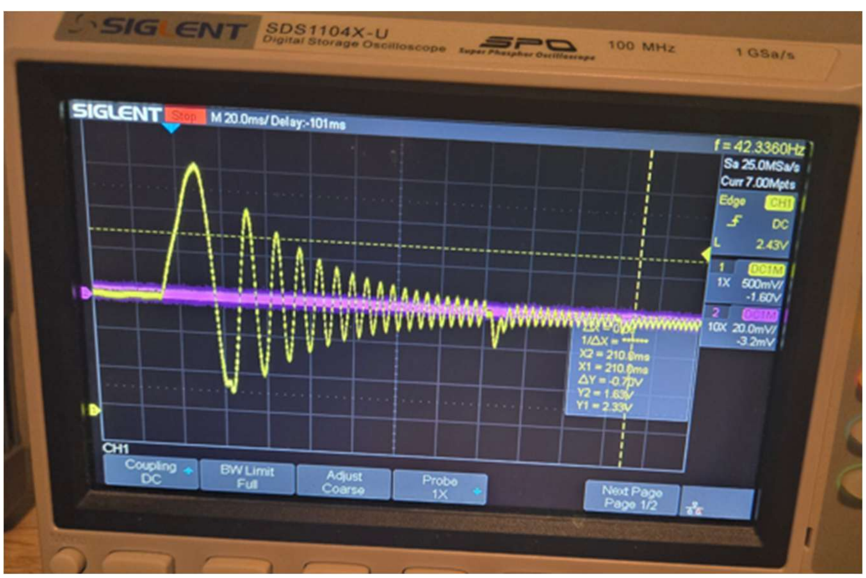
1st order



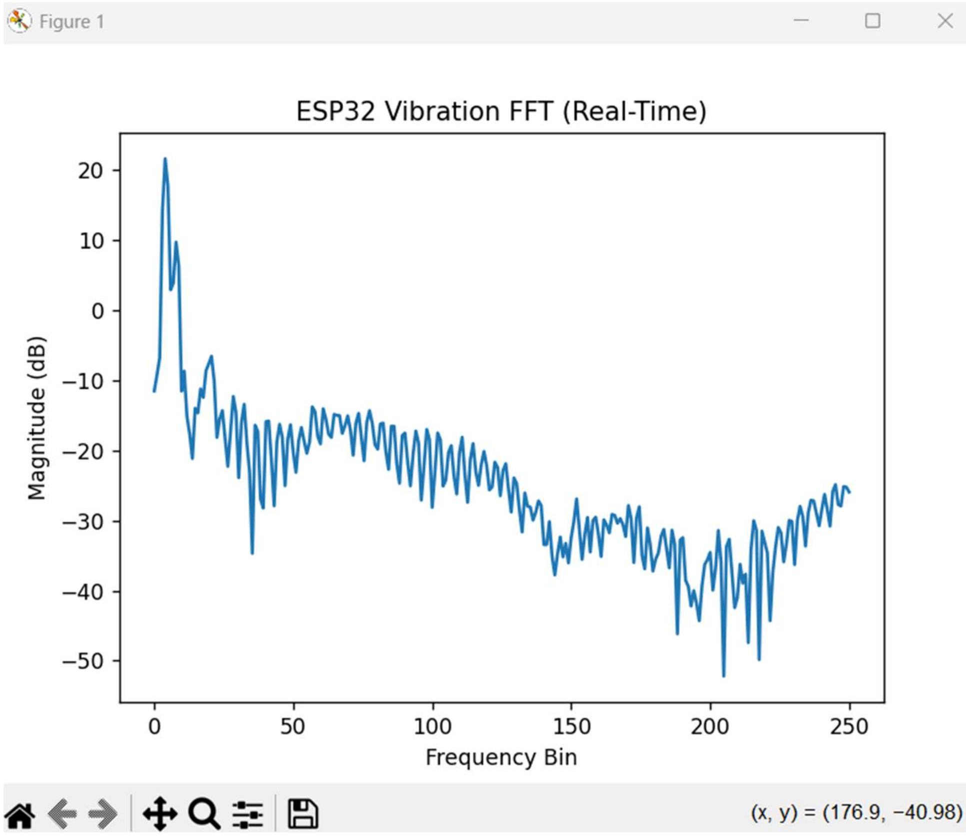
2nd Order



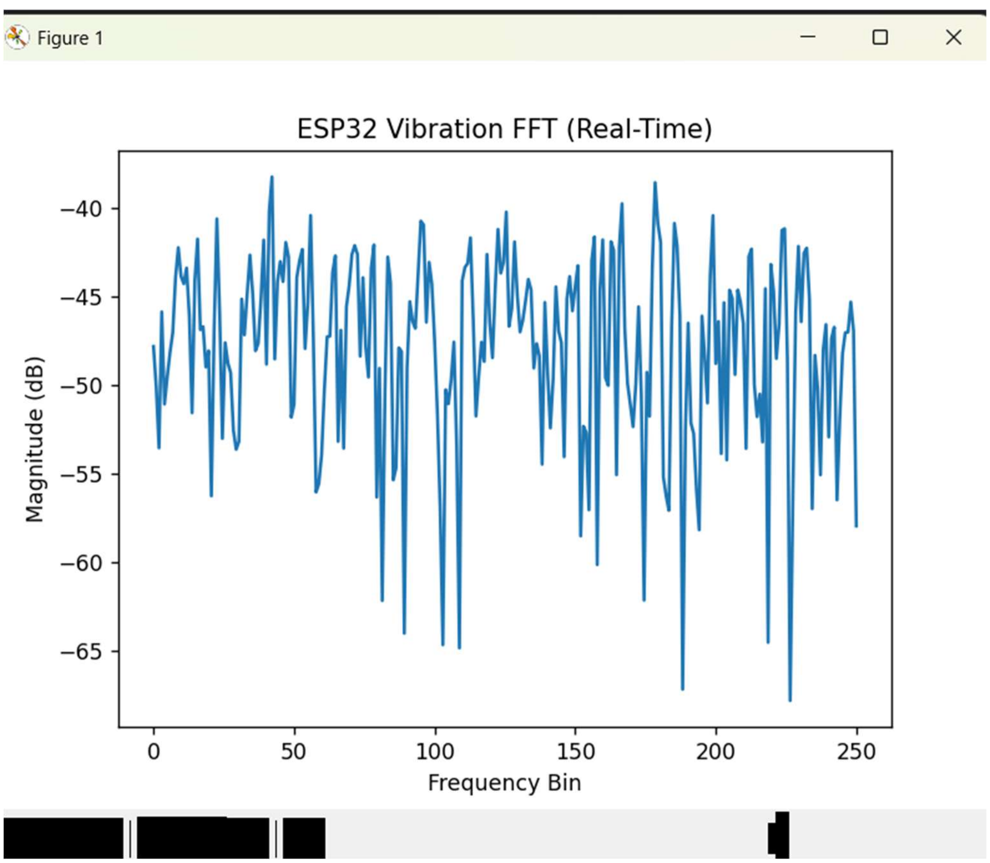
3rd order:



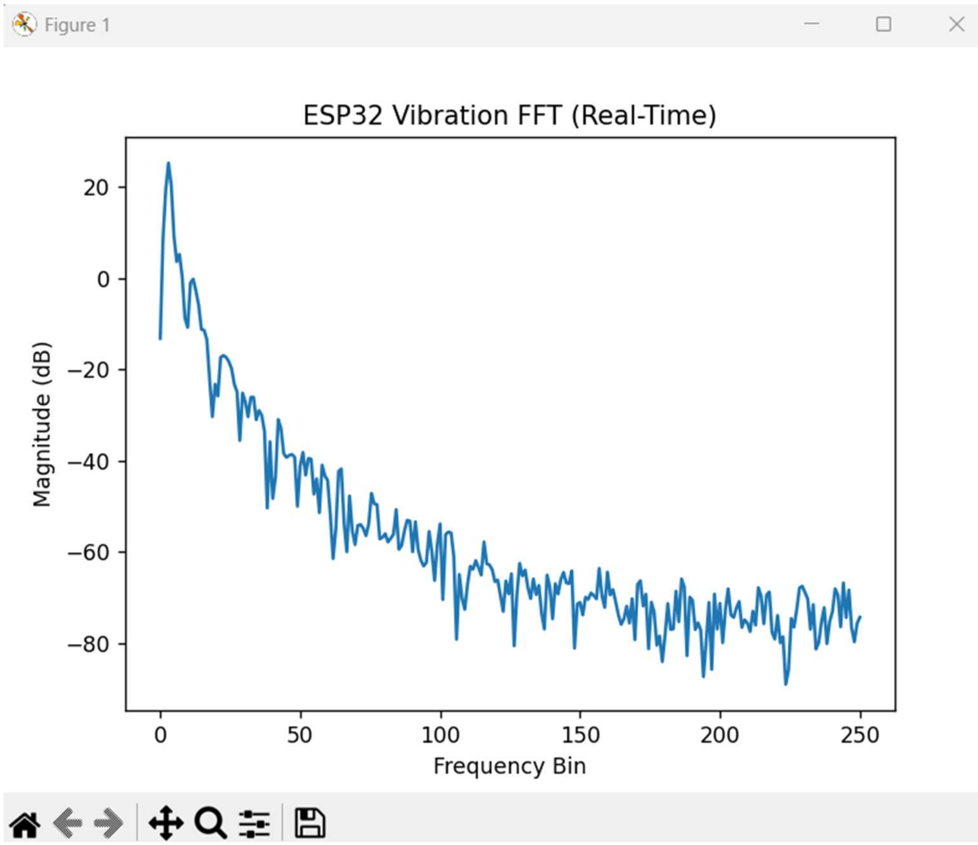
Accelerometer Z axis FFT (slow oscillation in z direction). Unfiltered.



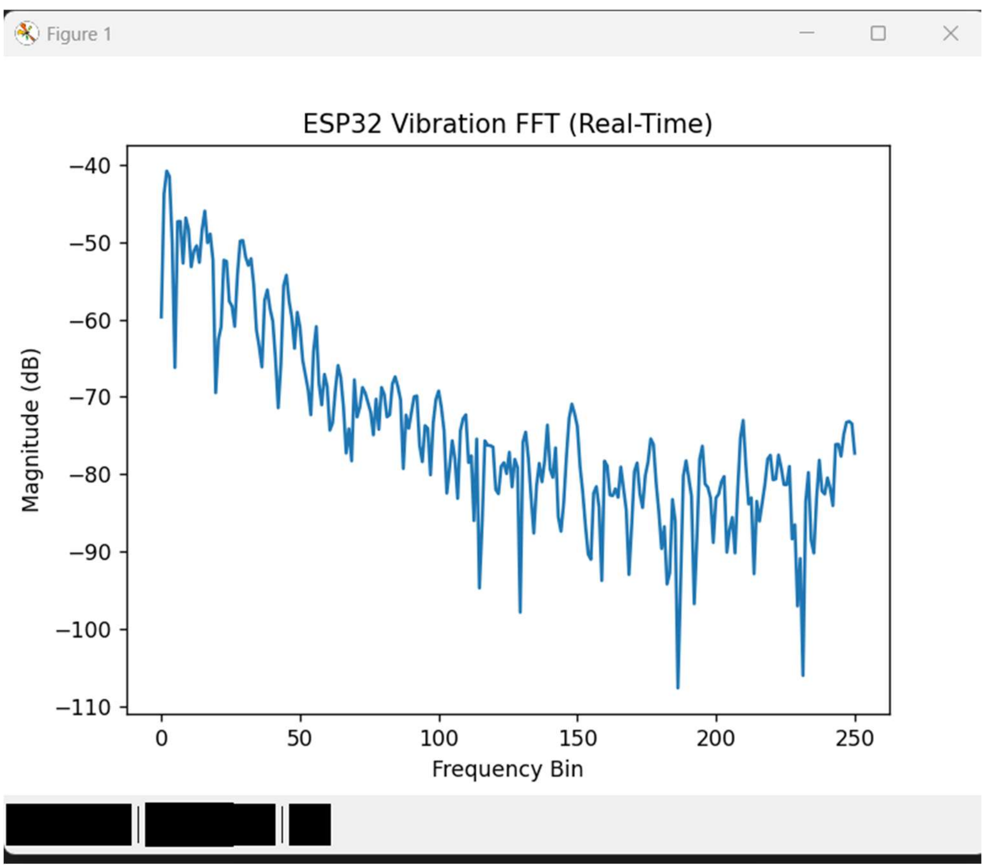
No oscillation:



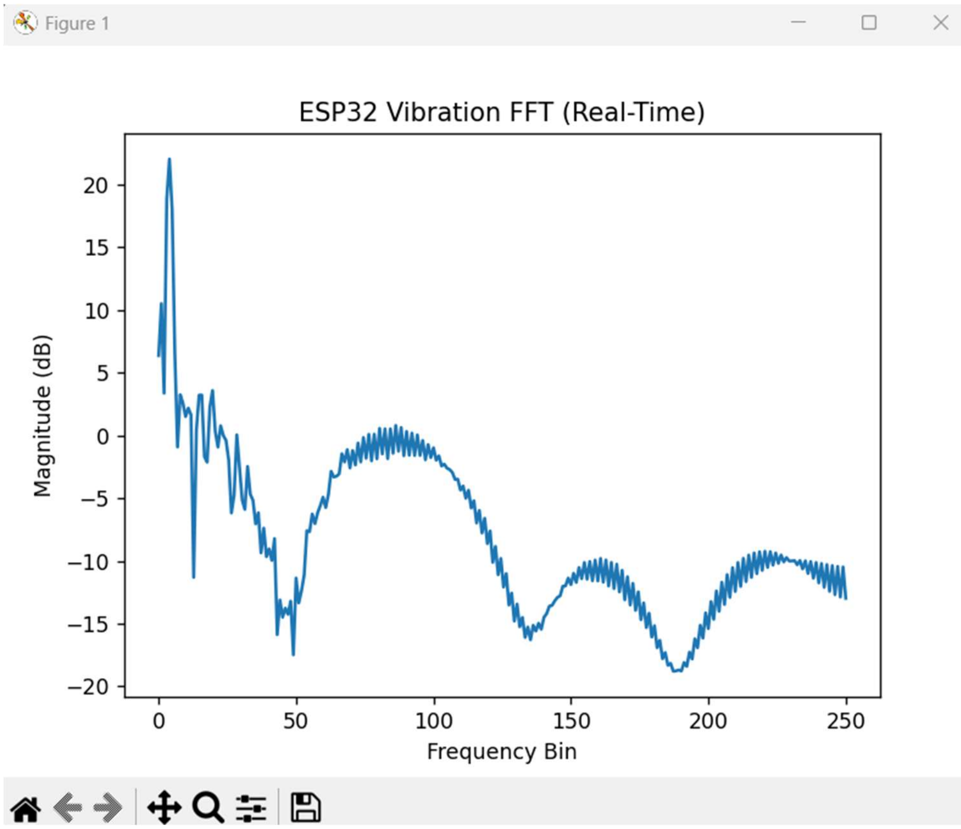
Filtered by second order EMA with 31 Hz cutoff, slow (~5 Hz) oscillation:



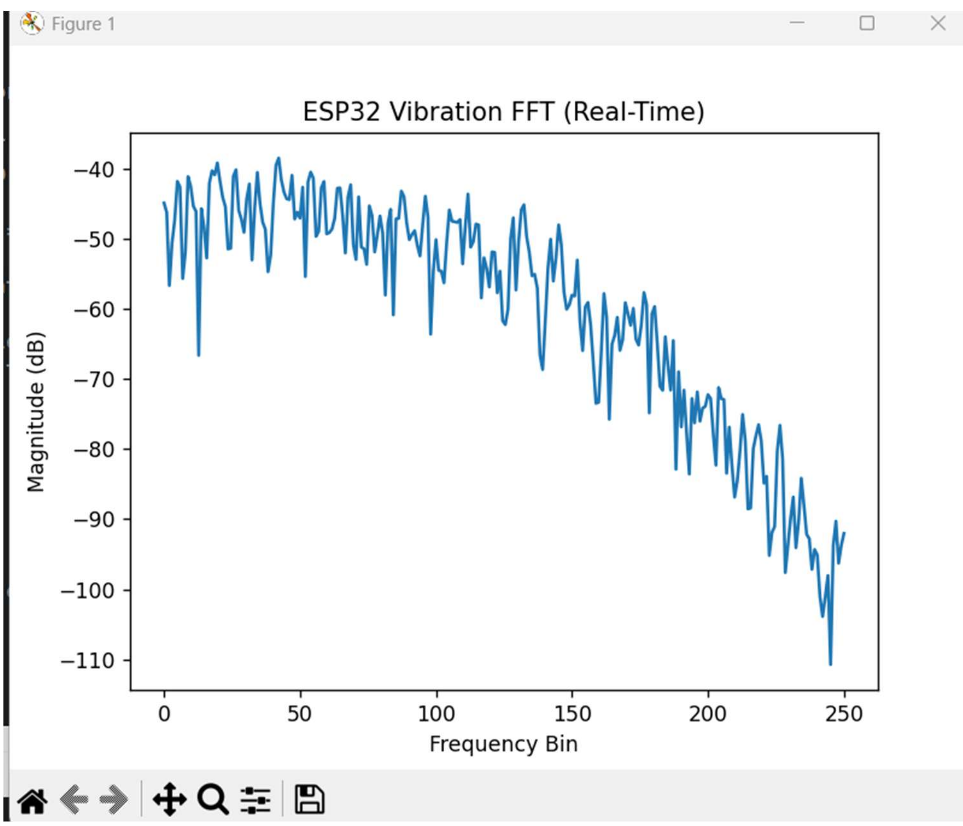
No oscillations:



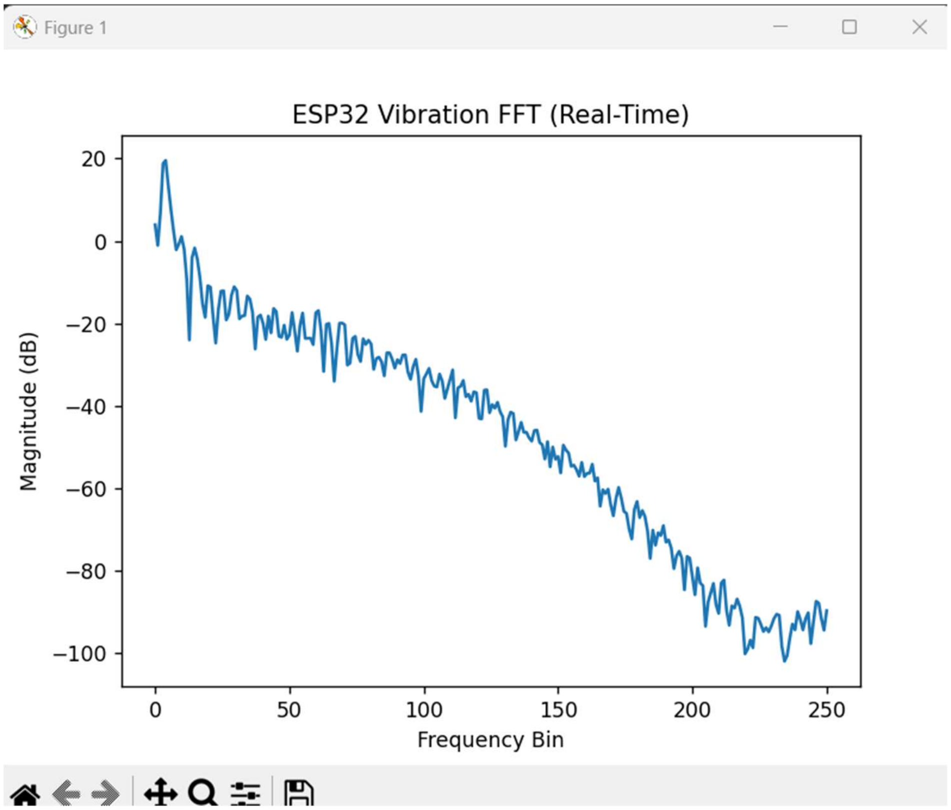
Filtered by second order Biquad with 100 Hz cutoff, slow (~5 Hz) oscillation:



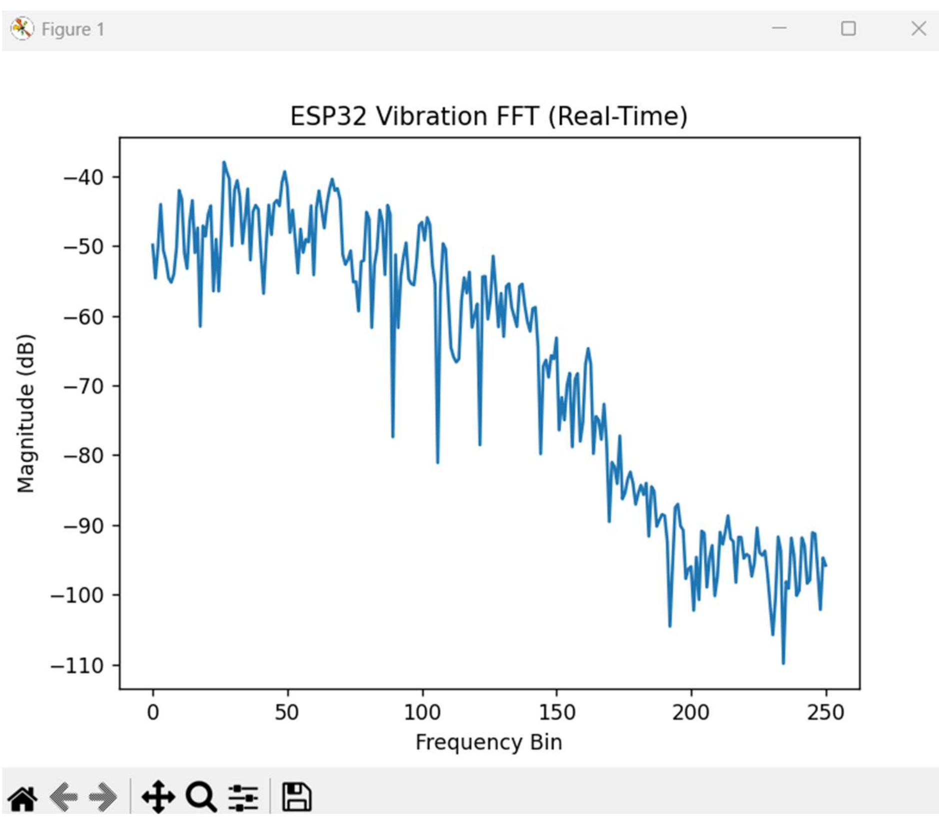
No oscillations:



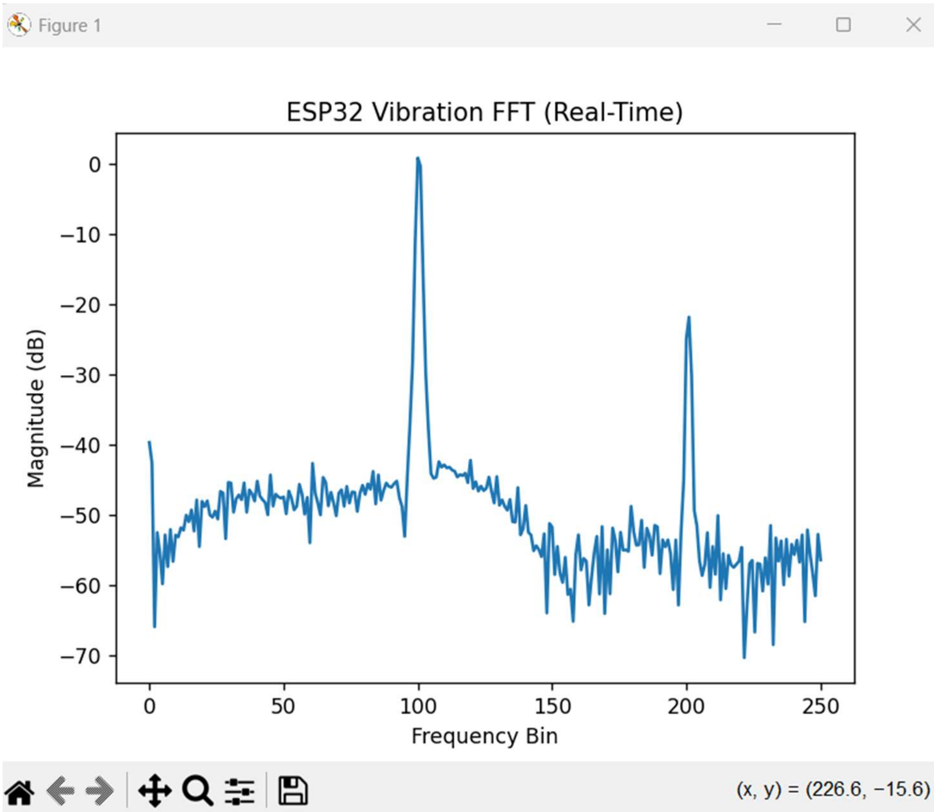
Filtered by fourth order Biquad with 100 Hz cutoff, slow (~5 Hz) oscillation:



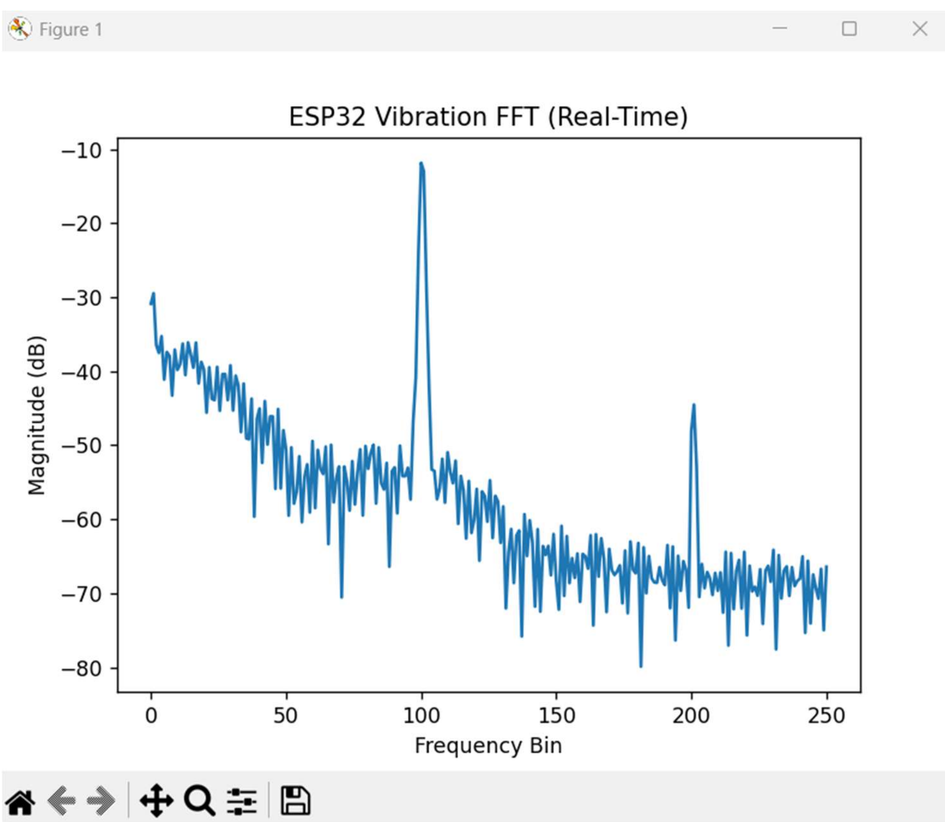
No oscillations:



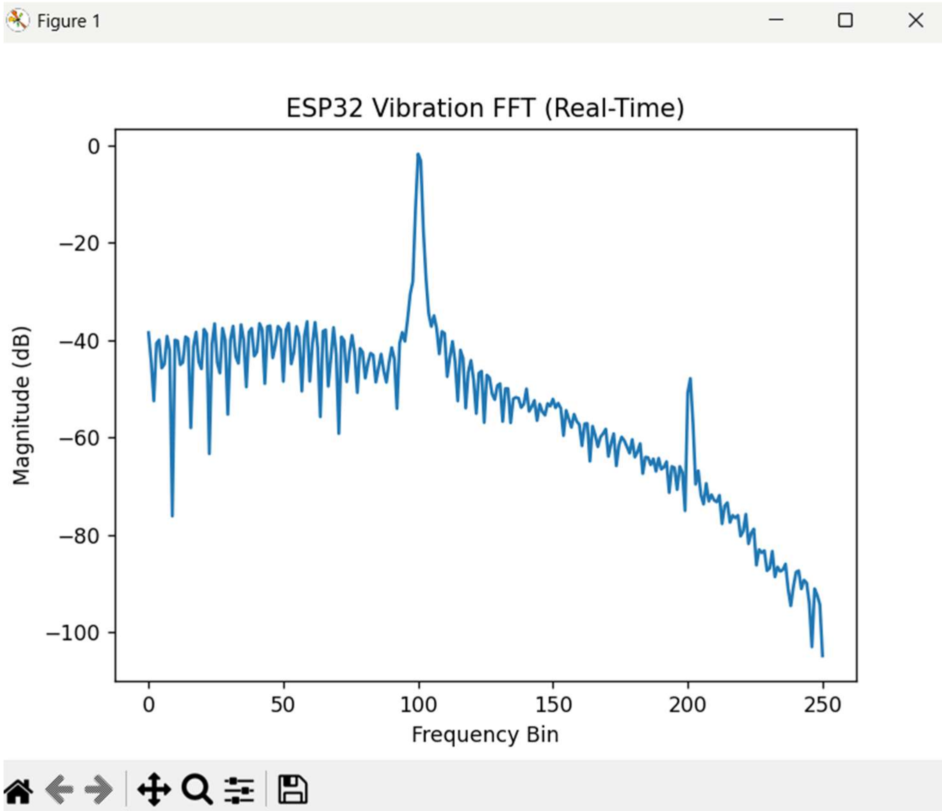
No filter. 100 Hz ADC input. 3.3 amplitude, 1.550 v offset



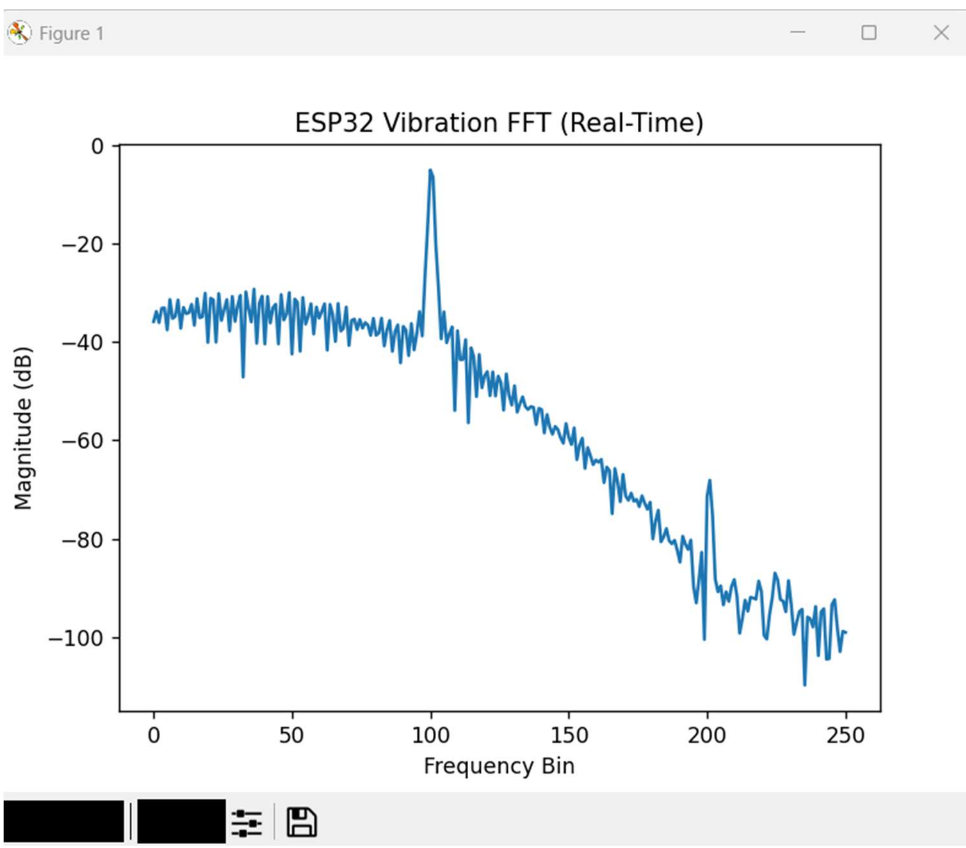
2nd order EMA, 100 Hz cutoff, alpha = 0.46. 100 Hz ADC input. 3.3 amplitude, 1.550 v offset.



2nd order Biquad, 100 Hz cutoff, alpha = 0.46. 100 Hz ADC input. 3.3 amplitude, 1.550 v offset.



4th order Biquad, 100 Hz cutoff, alpha = 0.46. 100 Hz ADC input. 3.3 amplitude, 1.550 v offset.



4^{th} order Biquad, 100 Hz cutoff, alpha = 0.46. 100 Hz AM at 50 Hz ADC input. 3.3 amplitude, 1.550 v offset.